Speech Enhancement Benchmark Lab — Report

All numbers come from runs in this repository (`results/r1`, `results/r2`, `results/latency.json`). Nothing here is estimated unless it says so. Machine: 2 vCPU aarch64, no GPU.

1. Executive summary

- **No method won everywhere.** The best method depends on the kind of interference, and the metric changes the answer.
- **Competing talker: target-speaker extraction (TSE) is the only single-channel method that clearly helped.** The method is Conv-TasNet separation, then picking the stream whose ECAPA speaker embedding matches a ≈10 s enrolment clip of the target. Results:
  - SI-SDR +16.4 dB in round 1 and +15.4 dB in round 2.
  - ESTOI +0.30 (r1) and +0.28 (r2).
  - ASR word-correct rate +0.31 (r1) and +0.43 (r2).
  - Every enhancement-only model *degraded* this condition: Demucs −3.7 dB, MetricGAN+ −4.0 dB.
- **Stationary or non-speech noise: neural denoisers win on SI-SDR, ESTOI and PESQ.** This covers pink, traffic composite and broadband environmental noise. Demucs DNS64 gained +9.6 to +12.2 dB SI-SDR and +0.18 to +0.24 ESTOI across the two rounds. Classical methods gained only +1–4 dB and $\leq$ +0.08 ESTOI.
- **That objective gain did not carry over to ASR intelligibility.** On those same conditions, Whisper's word-correct rate on Demucs output was unchanged or slightly lower (Δ −0.07 to 0.00). This matches the literature (Hu & Loizou 2007): single-channel enhancement raises SNR-type metrics much more readily than it raises intelligibility.
- **Babble (6 talkers) and reverberant babble are still largely unsolved.**
  - Babble: the best single-channel ESTOI gain was +0.13 (TSE). TSE's ASR word-correct rate moved −0.03 in round 1 and +0.13 in round 2, so the effect was not stable.
  - Reverberant babble: every method landed between −0.09 and +0.04 ESTOI of the unprocessed signal. Most neural models *lowered* SI-SDR in round 2 (Demucs −3.6 dB).
- **In the simulated 4-mic array, cheap spatial filtering gave the largest ASR gains, at a fraction of the compute.**
  - Free-field MVDR with a blind noise mask, at RTF 0.009: word-correct rate +0.35 in babble and +0.26 with a competing talker.
  - DNN-mask MVDR, at RTF 0.39: only +0.15 and +0.04.
  - Adding a Demucs post-filter after the beamformer removed the ASR gain (−0.04 to 0.00).
  - Oracle MVDR (upper bound) reached +0.46 and +0.30.
- **Noise estimation is the bottleneck for classical methods.** Giving Wiener/LSA/spectral subtraction the true noise PSD (round 2) raised SI-SDRi from about +1.6 to +6.3 dB. ESTOI still rose only about +0.04, so even a perfect noise PSD buys little intelligibility.
- **MetricGAN+ games its training metric.** PESQ +0.43–0.48, but SI-SDR −3.5 to −4.0 dB, and ASR word-correct rate −0.15 to −0.19.
- **Offline speed does not mean it can stream.** Demucs DNS64 runs at RTF 0.39 offline. Run frame by frame through its own streamer it was *not* real-time on this CPU: RTF 2.52 with a 41 ms buffer, or 1.05 with an 89 ms buffer. Classical STFT methods need 32 ms of algorithmic latency and <0.1 ms of compute per 8 ms hop.

2. What was built

- `selab/` package. Details in README.md.
  - Seeded mixture builder.
  - 4 classical enhancers sharing one blind noise tracker, plus oracle-noise controls.
  - 4 pretrained neural models, plus TSE with ECAPA and oracle stream selection, and a hybrid denoise→TSE.
  - 5 beamformers.
  - Metrics: SI-SDR(i), STOI/ESTOI, PESQ-WB, Whisper base.en WER and word-correct rate.
  - Runtime, RTF, peak RSS and streaming latency.
  - Paired bootstrap analysis.

- **Round 1 (r1)**:
  - Single-channel: 6 conditions × SNR {−5, 0, +5} dB × 8 utterances.
  - 4-mic array: 2 conditions × 3 SNRs × 8.
  - 192 mixtures, 3–8 s each.
- **Round 2 (r2)**:
  - New target utterances, none used in r1, built with a different seed.
  - SNR range widened to {−10, 0, +10} dB, 10 utterances per cell, 180 mixtures.
  - Adds oracle-noise controls and the hybrid.
- Demo app (`app.py`, Gradio). Upload a file, run any of 8 methods, listen, see waveforms and spectrograms, get metrics when a clean reference is uploaded, and export the WAVs. It also browses every benchmark mixture and output with its measured metrics and ASR transcript.

3. Results

3.1 Round 1, single-channel (n = 144 per method; SepFormer n = 51: 3 utterances per cell, pink +5 dB missing after a job timeout)

Paired deltas vs the unprocessed mixture. wcr is ASR word-correct rate on the first 3 utterances per cell (n = 54).

method	ΔSI-SDR dB	ΔESTOI	ΔPESQ	wcr	Δwcr	RTF (offline)	peak RSS MB	params
unprocessed	0	0	0	0.566	0	–	–	–
spectral subtraction	+1.54	+0.022	+0.03	–	–	0.002	269	–
Wiener (DD)	+1.71	+0.015	+0.06	0.540	−0.026	0.004	273	–
MMSE-LSA	+1.68	+0.018	+0.07	–	–	0.009	273	–
noisereduce	+0.14	+0.009	−0.03	–	–	0.006	300	–
MetricGAN+	−3.49	−0.004	+0.43	0.375	−0.191	0.043	315	1.9 M
Demucs DNS64	+5.63	+0.116	+0.24	0.510	−0.056	0.391	931	33.5 M
SepFormer-WHAM16k (subset)	+2.00	+0.090	+0.16	0.516	−0.027	3.10*	1520–1867	25.6 M
TSE Conv-TasNet+ECAPA	**+8.26**	**+0.176**	**+0.44**	0.584	+0.018	0.452	1168	27.2 M
TSE, oracle stream pick	+8.39	+0.178	+0.44	0.589	+0.023	(oracle)		

*SepFormer RTF 3.10 was measured idle (smoke run). The batch run measured 5.26 with another job competing for the CPU. 95% bootstrap CIs are in `results/r1/summary.md`. Two examples: TSE ΔESTOI [0.16, 0.20]; Wiener ΔESTOI [0.01, 0.02].

Per condition, r1 ΔESTOI (`results/r1/figs/heat_d_estoi.png`):

	babble	traffic	broadband	pink	competing	reverb+babble
Wiener DD	0.00	0.00	0.04	0.08	−0.02	0.00
Demucs	0.10	0.23	0.23	0.24	−0.10	−0.01
SepFormer	0.07	0.23	0.24	0.15	−0.03	−0.09
TSE ECAPA	0.13	0.21	0.21	0.17	**0.30**	0.03

3.2 Round 1, 4-mic array (n = 48; ASR n = 18 per method)

method	ΔSI-SDR	ΔESTOI	wcr	Δwcr	RTF
mic 0 unprocessed	0	0	0.453	0	–
Demucs on mic 0	−0.68	−0.018	0.414	−0.039	0.39
TSE on mic 0	+3.57	+0.062	0.469	+0.016	0.45
delay-and-sum	−0.42	+0.019	0.626	+0.173	0.021
MVDR free-field + blind mask	+1.12	+0.061	**0.757**	**+0.305**	**0.009**
MVDR, DNN (Demucs) mask	+0.57	+0.021	0.546	+0.093	0.39
MVDR DNN mask + Demucs post-filter	+0.05	+0.025	0.433	−0.020	0.77
MVDR oracle covariances (upper bound)	+3.88	+0.118	0.830	+0.377	0.043

3.3 Round 2 (held-out utterances, −10/0/+10 dB, n = 180; ASR n = 72)

method	ΔSI-SDR	ΔESTOI	ΔPESQ	wcr	Δwcr
Wiener DD (blind noise)	+1.63	+0.010	+0.09	0.540	−0.016
Wiener DD, **oracle noise PSD**	+6.29	+0.041	+0.28	0.585	+0.029
MMSE-LSA, oracle noise	+6.11	+0.045	+0.29	–	–
spectral subtraction, oracle noise	+5.24	+0.047	+0.17	–	–
MetricGAN+	−4.02	−0.004	+0.48	0.406	−0.150
Demucs DNS64	+4.11	+0.088	+0.36	0.542	−0.014
TSE Conv-TasNet+ECAPA	**+6.92**	**+0.143**	**+0.49**	**0.590**	**+0.034**
Demucs → TSE (hybrid)	+4.41	+0.093	+0.32	0.526	−0.030

By input SNR (r2, ASR word-correct rate):

SNR	unprocessed	Wiener	Wiener + oracle noise	Demucs	TSE
−10 dB	0.16	0.14	0.19	0.13	**0.25**
0 dB	0.61	0.61	**0.69**	0.62	0.66
+10 dB	**0.90**	0.87	0.87	0.87	0.85

At +10 dB every method *reduced* ASR accuracy slightly: processing artifacts cost more than the noise they removed.

3.4 Latency and real-time feasibility (`results/latency.json`, idle machine)

method	algorithmic latency	compute	real time on this CPU?
spectral subtraction / Wiener / LSA	32 ms window	0.015 / 0.026 / 0.074 ms per 8 ms hop	yes, by about 100×
Demucs DNS64, streaming, 1 frame	41.3 ms	40.4 ms per 16 ms stride (RTF 2.52)	**no**
Demucs DNS64, streaming, 4 frames	89.3 ms	66.9 ms per 64 ms stride (RTF 1.05)	**no** (borderline)
MetricGAN+, SepFormer, Conv-TasNet TSE	non-causal (whole utterance)	offline RTF 0.04 / 3.1 / 0.45	not streamable as released
MVDR (as implemented)	utterance-level covariances	RTF 0.009–0.04	offline; needs recursive covariance updates to stream

The review documents latency tolerances: about 20 ms for closed fittings and about 5–6 ms for open fittings (Stone & Moore). By that standard, none of the neural models here meets an open-fit budget even with infinite compute.

4. Surprises

1. **Objective gains and ASR gains disagree.** Demucs added +9.6 to +12.2 dB SI-SDR on pink, traffic and broadband noise, yet ASR word-correct rate did not improve. Whisper is noise-robust and is sensitive to enhancement artifacts. This is a known effect in the ASR literature, and it means ASR is a weak stand-in for human intelligibility in both directions.
2. **Delay-and-sum on a 12 cm array barely moved SI-SDR (−0.4 dB) but raised ASR word-correct rate by +0.17.** That suggests the ASR rewards attenuating the interfering *speech* stream, which also stops the competing talker's words being transcribed, in ways SI-SDR does not capture.
3. **The beamformer with a DNN post-filter did worse than the beamformer alone on ASR.** Stacking "the best of both" was not additive.
4. **The hybrid Demucs→TSE split by noise type.** On non-speech noise it beat TSE alone (pink: +10.6 vs +6.5 dB SI-SDR, +0.18 vs +0.14 ESTOI). With a competing talker it collapsed (+1.8 vs +15.4 dB), and in babble it lost too (−0.8 vs +2.9 dB). Likely cause (not tested): the denoiser keeps both voices but adds artifacts, which gives the separator a harder problem. Routing by interference type would beat any fixed cascade.
5. **ECAPA picked the same stream as the oracle in 95.8% of r1 mixtures.** The exceptions were mostly array-competing and reverberant cases, at 88% each. Selection error is not the limiting factor; separation quality is.

5. Critique as a skeptical reviewer, and what round 2 changed

concern	effect on conclusions	round-2 action
Classical methods may lose only because of a weak noise tracker	could understate classical	oracle-noise controls added. Gap confirmed as noise estimation (+4.6 dB SI-SDR), but ESTOI and ASR gains stay small
Round-1 utterances reused across analysis, risking overfitting of conclusions	inflated certainty	r2 uses new target utterances only (none from r1), a new seed, and 10 per cell
SNR range too narrow	misses failure regimes	r2 covers −10 and +10 dB. TSE's edge grows at −10 dB; all methods hurt slightly at +10 dB
Too few ASR samples	noisy WER	r2 ASR n = 72 per method. Word-correct rate added because WER is inflated by insertions (competing-talker words, hallucinations)

concern	effect on conclusions	round-2 action
In-domain advantage for TSE: Conv-TasNet was trained on Libri2Mix (LibriSpeech) and tested on LibriSpeech speakers	**likely optimistic for TSE**, especially competing-talker	**not fixed**: no out-of-corpus speech was available offline. Top next experiment
Babble and competing talkers also drawn from LibriSpeech	same bias	not fixed
Synthetic rooms and arrays, ideal mic matching, known target direction	beamforming optimistic (free-field steering used the true azimuth)	partially addressed: MVDR-ff estimated noise blindly, but steering is still oracle-direction
ASR proxy is Whisper base.en, which may share training data with LibriSpeech	absolute WER optimistic; deltas more trustworthy	flagged
No loudness normalisation before listening	perceptual bias in the listening pack	flagged in the listening page
CPU contention during some batch runs	some RTFs inflated	idle measurements used where available (SepFormer smoke, latency.json). r2 RTFs not used for comparisons
PESQ applied to separated speech and low-SNR inputs where it is poorly validated	PESQ deltas less trustworthy than ESTOI	reported, but not used to rank

6. Limitations and unresolved questions

- No human listening test was run. Whisper word-correct rate is a proxy, and the methods above show that proxy and objective metrics can diverge.
- Single corpus (LibriSpeech read speech, 26 speakers in total), synthetic noise mixing, and simulated rooms.
- The 'traffic' condition is an ESC-50 composite: engine/train bed plus horns and sirens. It is not a recorded street.
- No conversational, Lombard or far-field real recordings. No real multichannel data.
- DeepFilterNet and RNNoise could not be built in rounds 1-2. Both were built and benchmarked in round 3 (§8).
- Every processed signal here is digital output. None of it says anything about acoustic noise reaching a listener's ear (passive, ANC or leakage); see review §8.
- Open: does TSE's competing-talker advantage survive out-of-domain speakers and real rooms? Does any method raise *human* keyword recognition in babble?

7. Recommended next experiments (highest information value first)

1. **Out-of-domain stress test for TSE and Demucs.** Use a non-LibriSpeech corpus (e.g. VCTK or conversational speech) with *recorded* noise (DEMAND or WHAM!), keeping the same pipeline. This decides whether the headline TSE result is real or an artifact of matching training and test data.
2. **A small human keyword-recognition test.** Use a randomised, blinded web form with matrix sentences or keyword scoring, built from the existing listening pack. Unprocessed vs Wiener vs Demucs vs TSE vs MVDR-ff, at the SNRs where the proxies disagree (0 dB babble, 0 dB pink). It is the only way to settle whether the ESTOI gains or the ASR results reflect real listeners.
3. **Streaming-constrained comparison.** Implement recursive-covariance MVDR plus a causal mask, and test DeepFilterNet or another low-latency causal DNN on a machine that can build it. Score everything at a fixed latency budget (e.g. $\leq$10 ms and $\leq$40 ms), on real multi-mic recordings with an estimated, not known, target direction. Today the cheapest method (MVDR-ff) looks best per unit of compute, but only under oracle steering and offline covariances.

8. Round 3: executing the recommended experiments (2026-10-15)

All numbers are measured. Full tables are in `results/r3/summary_r3.md`, `generalisation_gap.csv` and `latency_budget.csv`; the run and its failures are in EXECUTION_LOG.md items 16-25. Reproduce with `scripts/run_round3.sh` followed by `python -m selab.analyze_r3`.

New this round. RNNoise (pyrnnoise wheel; its 20 ms delay is measured and compensated) and DeepFilterNet3 (built from source with a user-space Rust toolchain and zig as the linker). Both also ran on r1 and r2. Also new: an out-of-domain set (r3), true hop-by-hop streaming implementations, and a blinded human listening-test app.

8.1 Out-of-domain test (144 mixtures: CMU ARCTIC talkers incl. Scottish- and Indian-English targets; recorded MS-SNSD noise)

method	ΔSI-SDR dB	ΔESTOI [95% CI]	ΔPESQ	Δ ASR word-correct [95% CI]	RTF
TSE Conv-TasNet + ECAPA	+10.2	+0.199 [0.177, 0.222]	+0.54	**+0.11 [0.04, 0.17]**	0.91
Demucs DNS64	+7.1	+0.134 [0.107, 0.161]	+0.38	-0.02 [-0.06, 0.02]	0.61
DeepFilterNet3	+4.9	+0.103 [0.083, 0.122]	+0.36	-0.05 [-0.09, -0.01]	0.088
RNNoise	+3.1	+0.069 [0.050, 0.088]	+0.15	-0.09 [-0.14, -0.05]	0.092
Wiener (DD)	+1.4	+0.009 [0.003, 0.014]	+0.03	-0.02 [-0.05, 0.00]	0.004
MetricGAN+	-4.3	+0.009 [-0.005, 0.022]	+0.43	-0.12 [-0.17, -0.07]	0.04

- **The headline TSE result survives the domain shift.** Competing talker OOD: +16.2 dB SI-SDR, +0.33 ESTOI, +0.54 ASR word-correct rate, against +15.8 dB / +0.29 in-domain. The ECAPA-selected output matches the oracle-selected one to within 0.4 dB overall. It is the only method whose ASR gain is positive with a CI excluding zero.
- **No method collapsed out of domain.** On recorded babble, every neural method gained *more* than on the in-domain LibriSpeech-built babble (Demucs ΔESTOI +0.26 vs +0.07). That is consistent with recorded crowd babble being less target-like than babble built from the same corpus as the target. It is a hypothesis, not tested.
- **Objective/ASR disagreement replicates out of domain.** DeepFilterNet3 and RNNoise raise ESTOI on every recorded-noise condition but *lower* ASR word-correct rate overall. Reverberant OOD babble stays unsolved (all methods within -0.10 to +0.04 ESTOI).
- **RNNoise harms babble, reverberation and competing speech** in every round (r1 babble ΔESTOI -0.14, reverb_babble -0.18). Its benefit is limited to stationary or quasi-stationary noise.

8.2 Latency-budget comparison (algorithmic latency = window + lookahead)

budget	best measured option	ΔESTOI	Δ ASR wcr	compute
≤10 ms	streaming Wiener, 8 ms window	-0.003 (no benefit)	n/a	0.055 ms per 4 ms hop
≤20 ms	RNNoise	+0.069 OOD (but -0.09 ASR)	-0.09	RTF 0.092
≤40 ms	DeepFilterNet3	+0.103 OOD	-0.05	RTF 0.088 (offline batch)

budget	best measured option	ΔESTOI	Δ ASR wcr	compute
≈41 ms	Demucs streaming	+0.134	-0.02	**RTF 2.52 at 1 frame: not real-time here**
32 ms, 4 mics (simulated)	online MVDR, free-field steering	+0.035	**+0.24** (offline MVDR +0.31)	0.23 ms per 16 ms hop
52 ms, 4 mics	online MVDR → RNNoise	-0.076	-0.10	0.23 ms per hop + RNNoise

- **Below ≈16 ms nothing single-channel helped.** The hop-by-hop Wiener filters with online minimum-statistics noise tracking gave slightly *negative* ESTOI on the OOD set. The offline Wiener with the same 32 ms window scored +0.009. The gap is the cost of tracking noise causally.
- **For a real-time CPU budget, DeepFilterNet3 is the best single-channel option measured:** 84-96% of Demucs's ESTOI gain on stationary and environmental noise (r1 pink, broadband, traffic; r3 HVAC, transients), at about 1/7 of Demucs's compute and with a 40 ms latency it can actually meet. Its frame-by-frame compute was not measured: the Python API is batch-only.
- **With multiple microphones, the online recursive MVDR keeps most of the offline ASR gain** (+0.24 vs +0.31 word-correct) at under 1.5% of one core. Adding RNNoise after it erases the gain, the same pattern as Demucs post-filtering in round 1.
- Caveat: the MVDR rows still use the *known* target azimuth and simulated rooms. Steering-error robustness and real recordings remain untested.

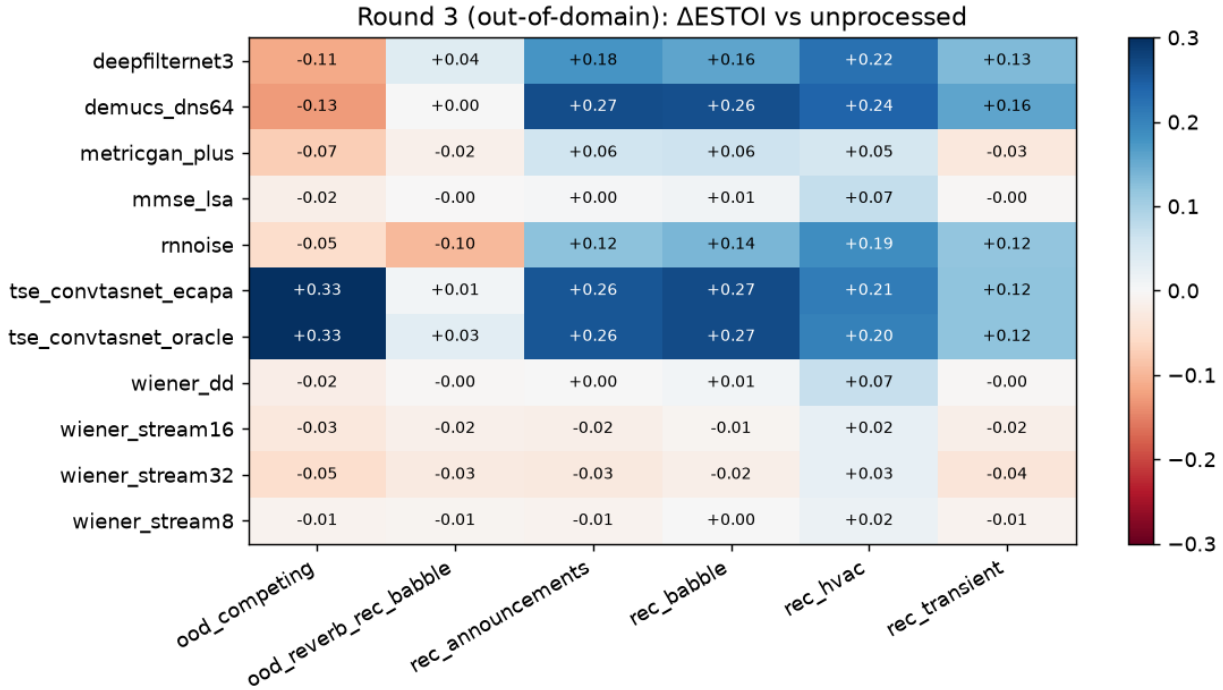


Figure 1: Round 3: ΔESTOI by out-of-domain condition

8.3 Blinded listening test: built, not run on people

`listening_test.py` serves a Gradio test. It draws 48 trials per listener from the OOD set (4 conditions × 0/-5 dB × 6 sentences). Six processing conditions are assigned by a cyclic Latin square, so each listener hears each sentence once and each method 8 times. Stimuli are RMS-equalised under opaque file names, playback is limited to one play, and listeners type the words they heard and rate listening effort. Scoring is the same word-correct rate as

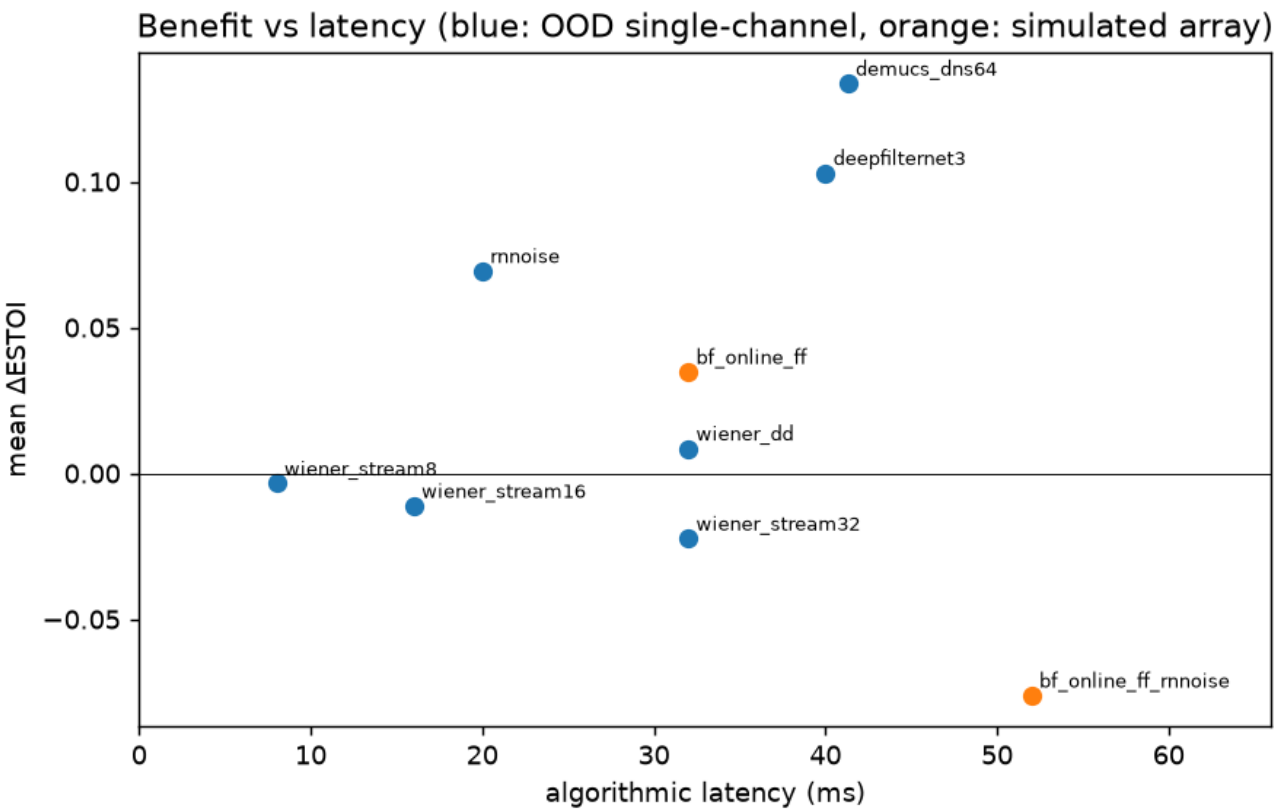


Figure 2: Benefit vs algorithmic latency

the ASR proxy, logged to CSV, and `--analyze` gives per-method bootstrap CIs. Verified: the app answers HTTP 200, the Latin-square balance was checked programmatically, and a synthetic-listener run (`--selftest`, Whisper standing in for a person, tagged `synthetic`) exercises the whole record → score → analyse path. **No human data exists yet.** Rough power: the trial-level SD of word-correct rate in the matching ASR data is about 0.36. With about 12 listeners (96 trials per method), the 95% CI half-width on a method mean is about ±0.07 (unpaired; paired within-sentence contrasts should be tighter). That is enough to detect the 0.1-0.5 differences seen for TSE in competing speech, but not the few-hundredths gaps between denoisers.

8.4 What changed in the conclusions

1. TSE's competing-talker advantage is not a LibriSpeech artifact (it was the main open question from §6).
2. A deployable real-time single-channel tier now exists in the comparison: DeepFilterNet3 at 40 ms. RNNoise belongs to stationary noise only.
3. For multi-mic capture, the cheapest causal method (online MVDR) remains the one that most consistently helps the ASR proxy. It should be paired with steering estimation, not a DNN post-filter.
4. Still open: human intelligibility (the test is ready to run), real multichannel recordings, estimated rather than known direction, and reverberant babble.